

Data Cleaning & Preparation – Change Log Book (Summary File)

Dataset: E Commerce Sales

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The dataset is now clean, standardized, and ready to load into Excel for analysis.

Every action I took solved a specific problem that would have negatively affected the business’s reporting, revenue tracking, or customer trust.

Summary of Changes & Their Business Impact:

Category	What I did	Why This Mattered (Impact)
Text Standardization	Trimmed extra spaces and applied “Capitalized Each Word” to columns like CustomerName, Product, Category, and OrderStatus.	Prevents inaccurate grouping in Pivot Tables. Without this, the sales team would think “Beauty”, “beauty”, and “BEAUTY” are three different categories. This fix ensures we see the correct total sales per category in one single row, saving hours of manual work.
Customer Privacy	Removed Email, Phone, and Notes columns entirely.	Protects the company from data breaches and legal risks. By removing personally identifiable information (PII), this dataset can now be safely shared with analysts, stakeholders, or external partners without violating privacy regulations or customer trust.
Duplicate Removal	Removed only exact duplicate rows (using all columns except OrderID) to keep multi-product orders intact.	Prevents inflated revenue and stock counts. If we had blindly removed duplicates based on OrderID, we would have deleted legitimate products (like a Laptop and a Mouse in the same order). This fix ensures our total sales and units sold are mathematically accurate, preventing costly business decisions based on double-counted or missing revenue.
Quantity Fix	Fixed negative typos for Delivered/Shipped orders, while keeping negative strictly for Returned orders.	Gives the Finance team a truthful view of returns vs. sales. A delivered order cannot have -3 units. By flipping these typos to +3, we now correctly calculate total units sold. Meanwhile, keeping

		returns negative ensures that refunds are accurately subtracted from total revenue.
Unit Price Fix	Stripped out currency symbols (NGN, \$, commas) and flipped negative prices to positive.	Unlocks the ability to calculate Total Revenue. Before this, mixing text (\$, NGN) with numbers made it impossible to do math. This fix allows us to accurately calculate $\text{Quantity} * \text{Unit Price} = \text{Revenue}$, giving the business a reliable metric for financial performance.
Data Standardization	Handled mixed US (MM/DD) and UK (DD/MM) date formats and converted them to a single Date type.	Enables accurate time-based analysis. With clean dates, we can now reliably analyze month-over-month sales growth, seasonal trends, and shipping delays (ShipDate - OrderDate) without dates being misread (e.g., Dec 1st being misread as Jan 12th).
Data Quality Flag	Created a separate "Quantity_Quality_Flag" column to track missing (null) values.	Creates transparency on data reliability. The business can now instantly see how many orders are missing a Quantity. Instead of hiding the problem, we highlight it so the team knows exactly how much of the data is trustworthy and where data entry training is needed most.
System Recommendations	Advised the business to use dropdown menus, mandatory fields, and date pickers in their data entry systems.	Saves the company hundreds of hours of future manual cleaning. Instead of fixing human errors repeatedly, these technical changes prevent errors from entering the system in the first place. This shifts the business from reactive (fixing data monthly) to proactive (generating clean data automatically).